ABSTRACT

Face Anti-Spoofing for Robust Multi-Attack Liveness Detection

Face recognition has become one of the most widely used biometric authentication methods in smartphones, banking systems, attendance management, and access control applications. However, these systems are vulnerable to spoofing attacks such as printed photographs, replay videos, mobile screen replays, and 3D masks, which can lead to unauthorized access. Therefore, an effective face anti-spoofing system is essential to verify whether the presented face is genuine or fake before authentication.

This project proposes a real-time face anti-spoofing system using deep learning to detect spoof attacks through a standard RGB webcam. The system captures live face images, detects the face using MediaPipe or YOLO Face, preprocesses the detected face, and extracts facial features using the EfficientNet-B0 model. The extracted features are classified to determine whether the face is live or spoof. The solution is implemented as a web application using React for the frontend and Flask for the backend, enabling real-time prediction and authentication.

The proposed system aims to provide a lightweight, cost-effective, and accurate anti-spoofing solution without requiring additional hardware such as depth sensors. In future work, the system can be extended to classify different spoof attack types, including printed photo attacks, replay video attacks, mobile screen replay attacks, and 3D mask attacks, thereby improving the security and reliability of biometric authentication systems.

Keywords: Face Anti-Spoofing, Deep Learning, EfficientNet-B0, Face Recognition, Liveness Detection, MediaPipe, React, Flask, Biometric Security.